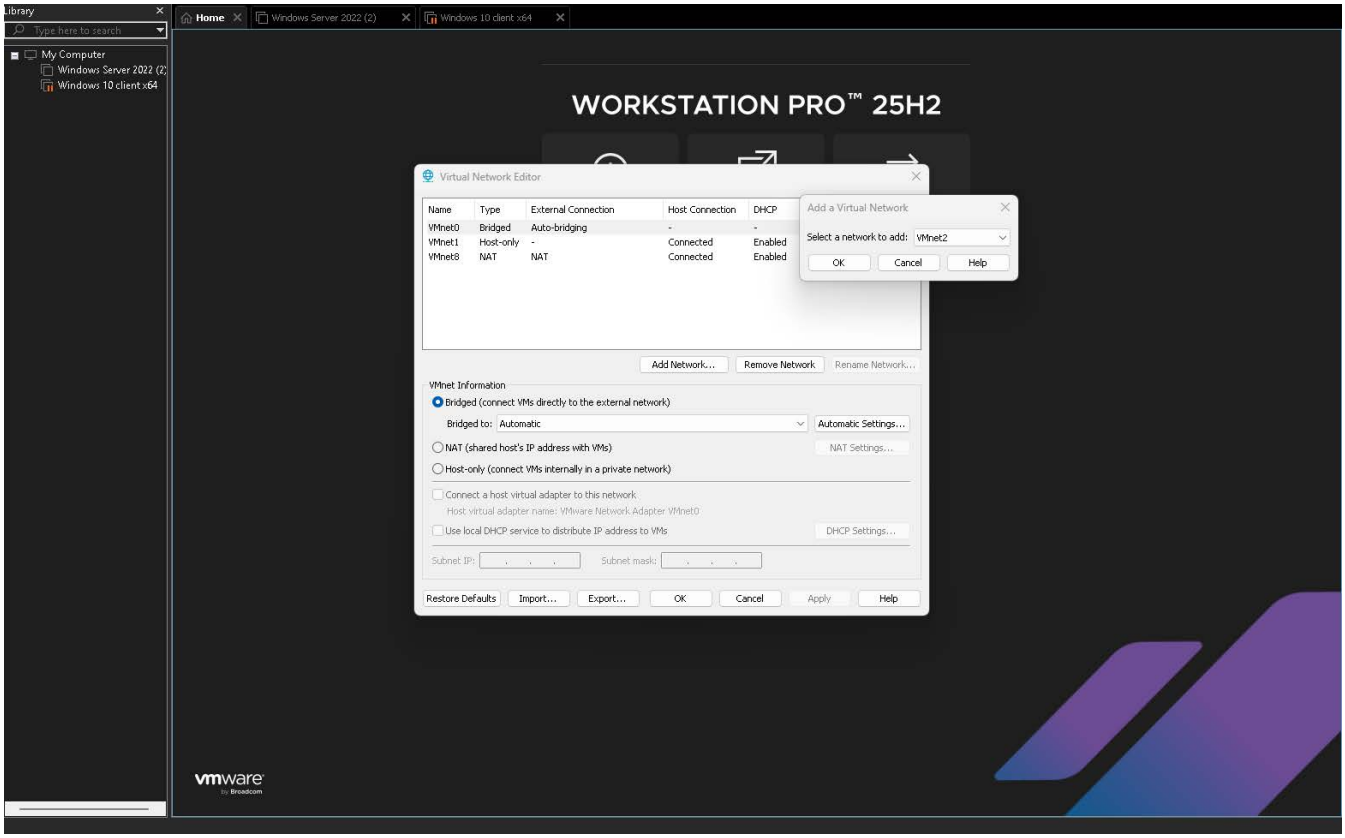
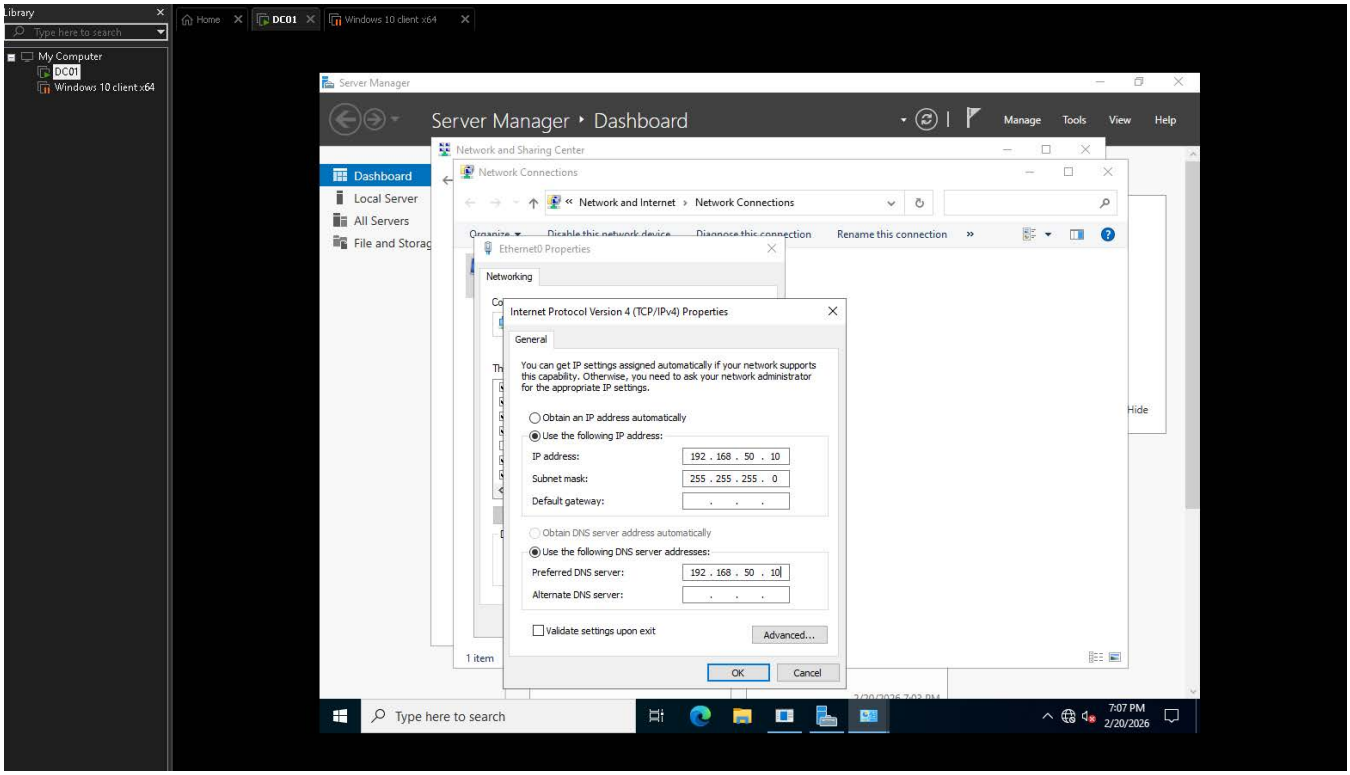
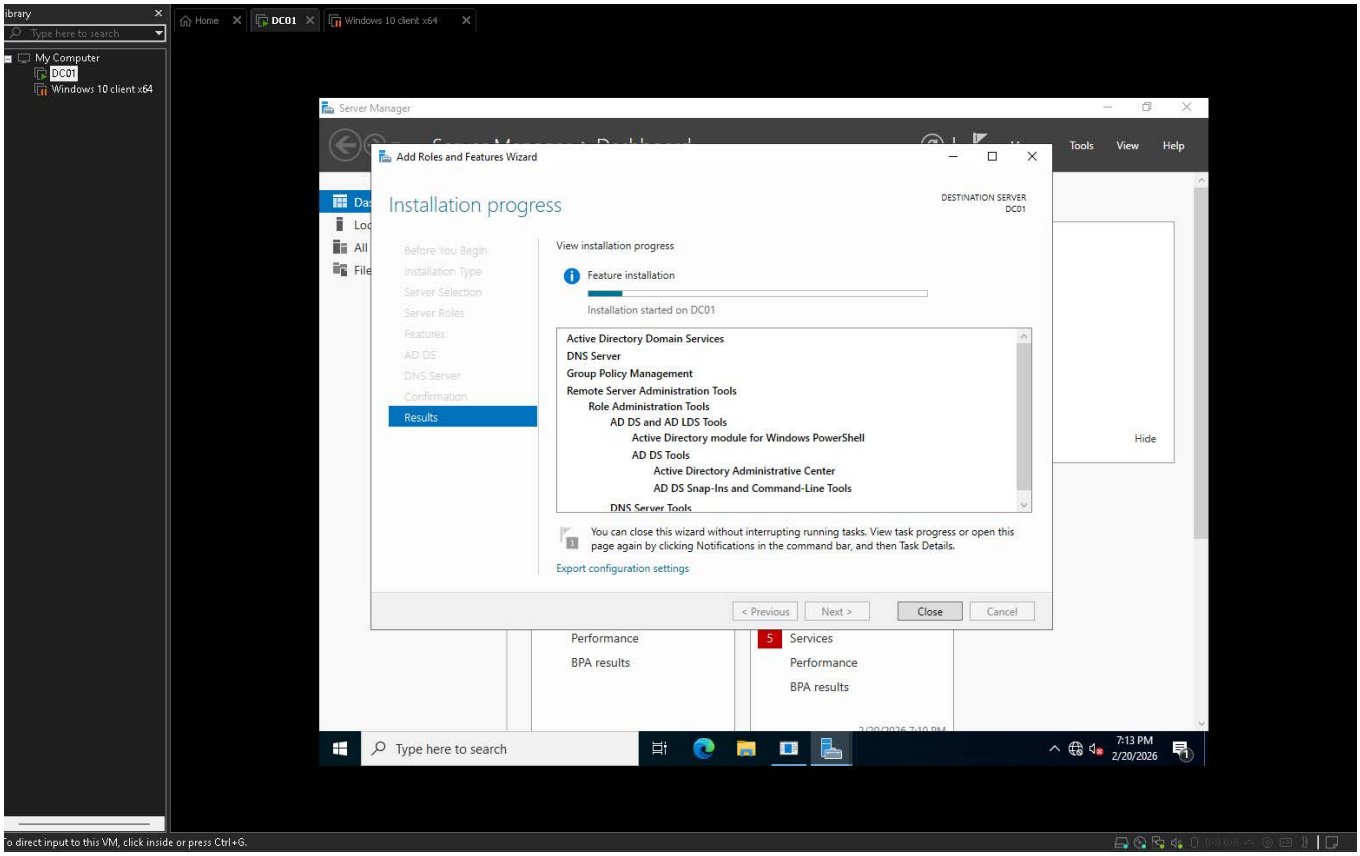
1. Creating internal network in VMware workstation pro.



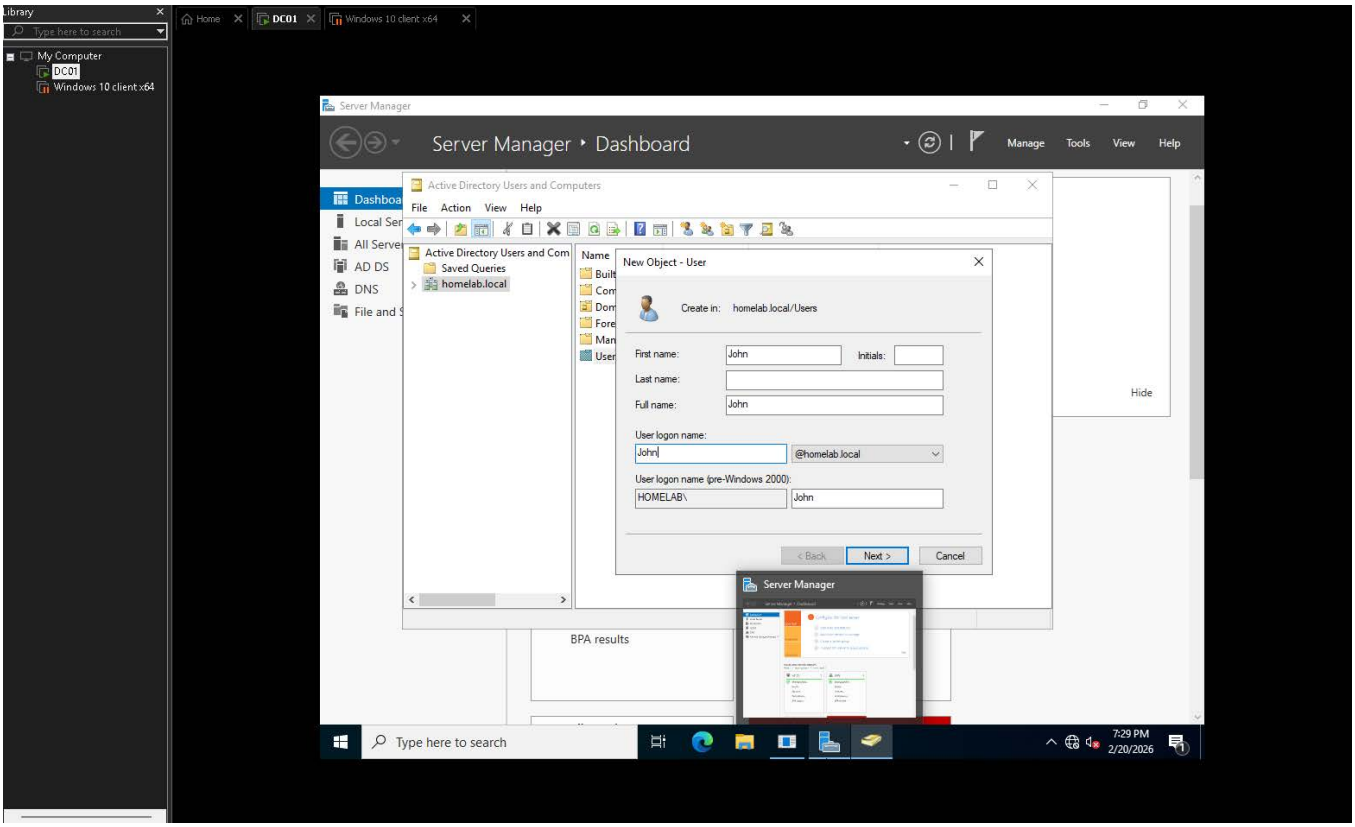
2. Configured static IP for DC01



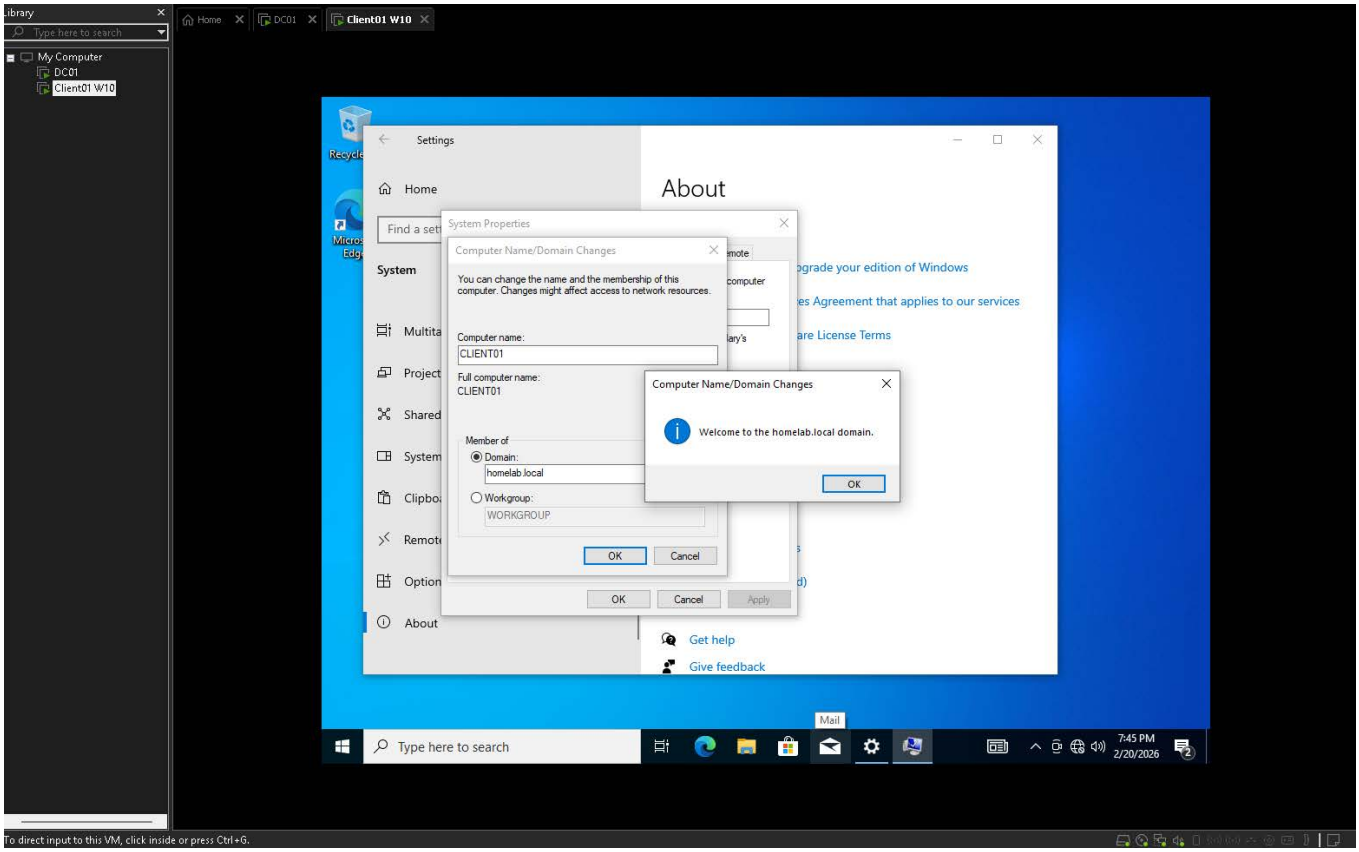
3. Installing AD DS and DNS server



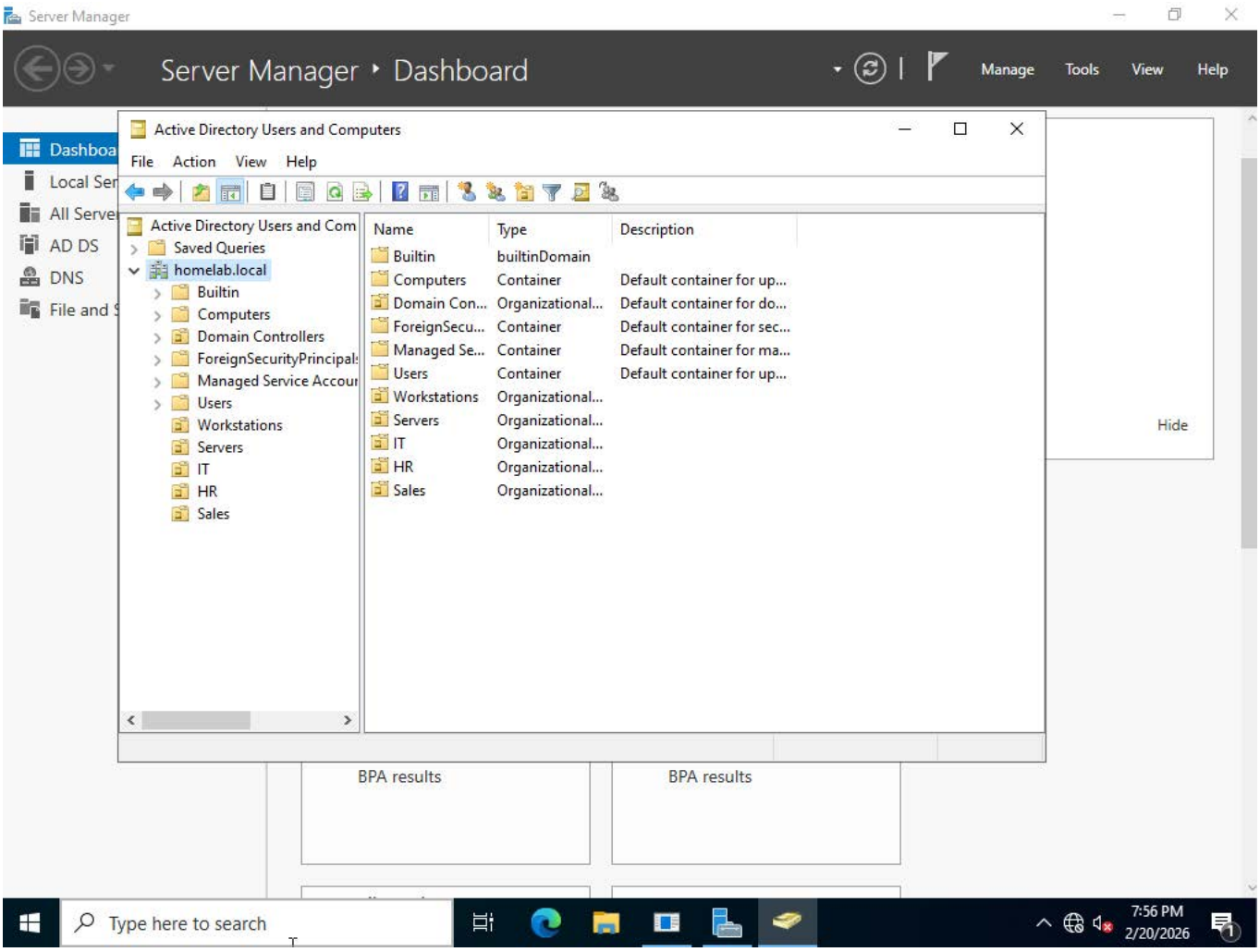
4. Created user John



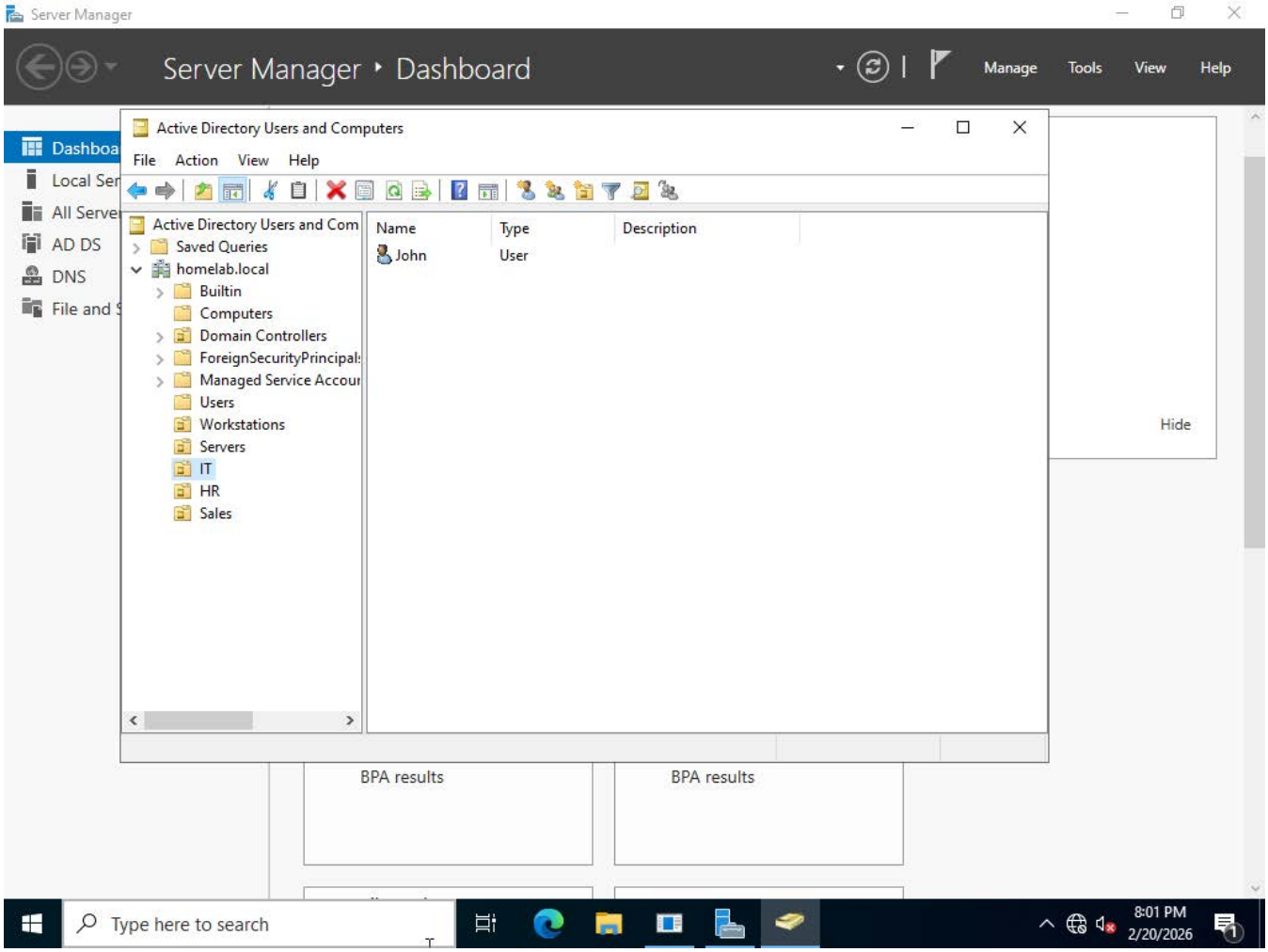
5. Created a client machine and joined it to domain homelab.local



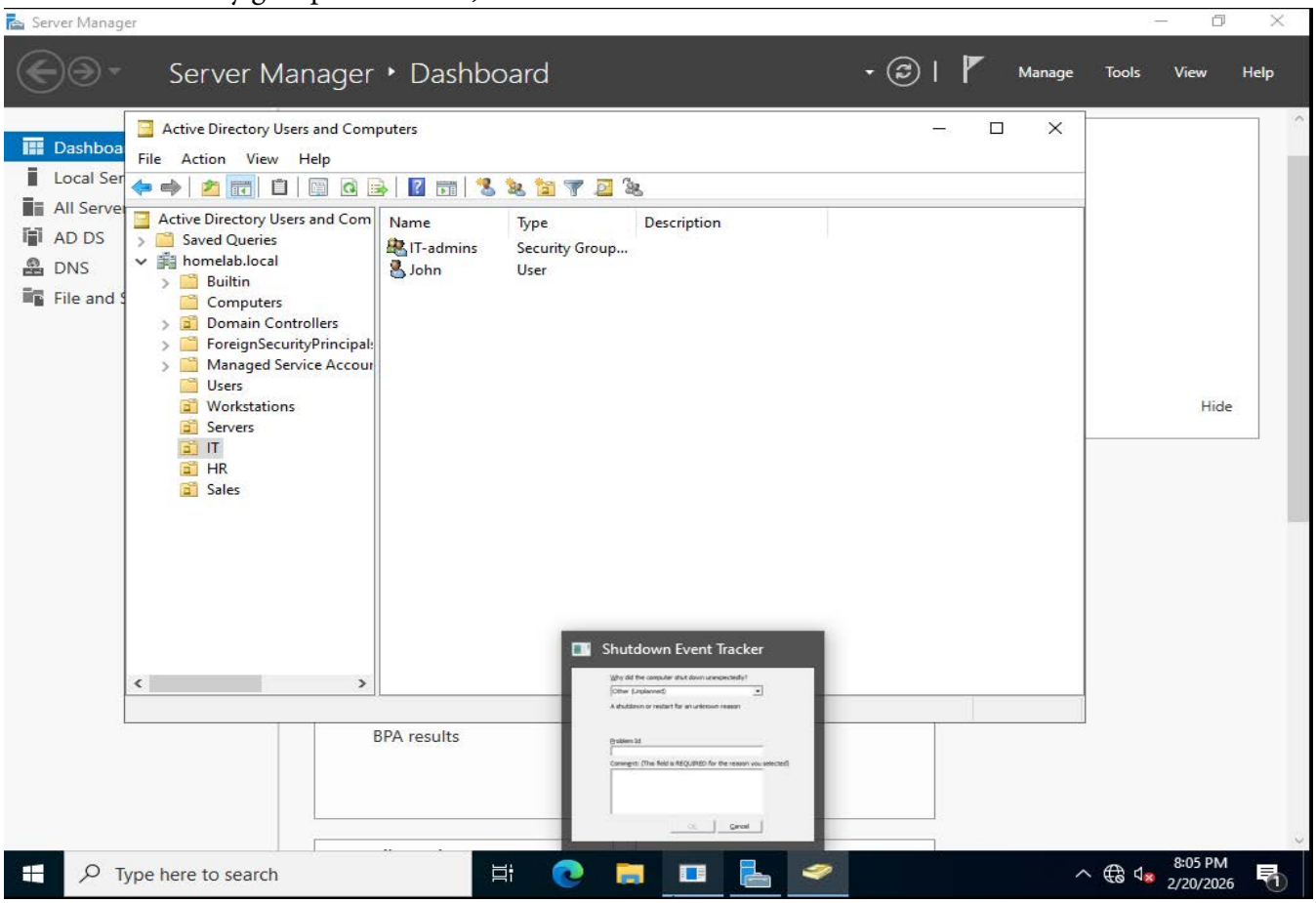
6. Created 5 OU's. workstations, Servers, IT, HR, and Sales



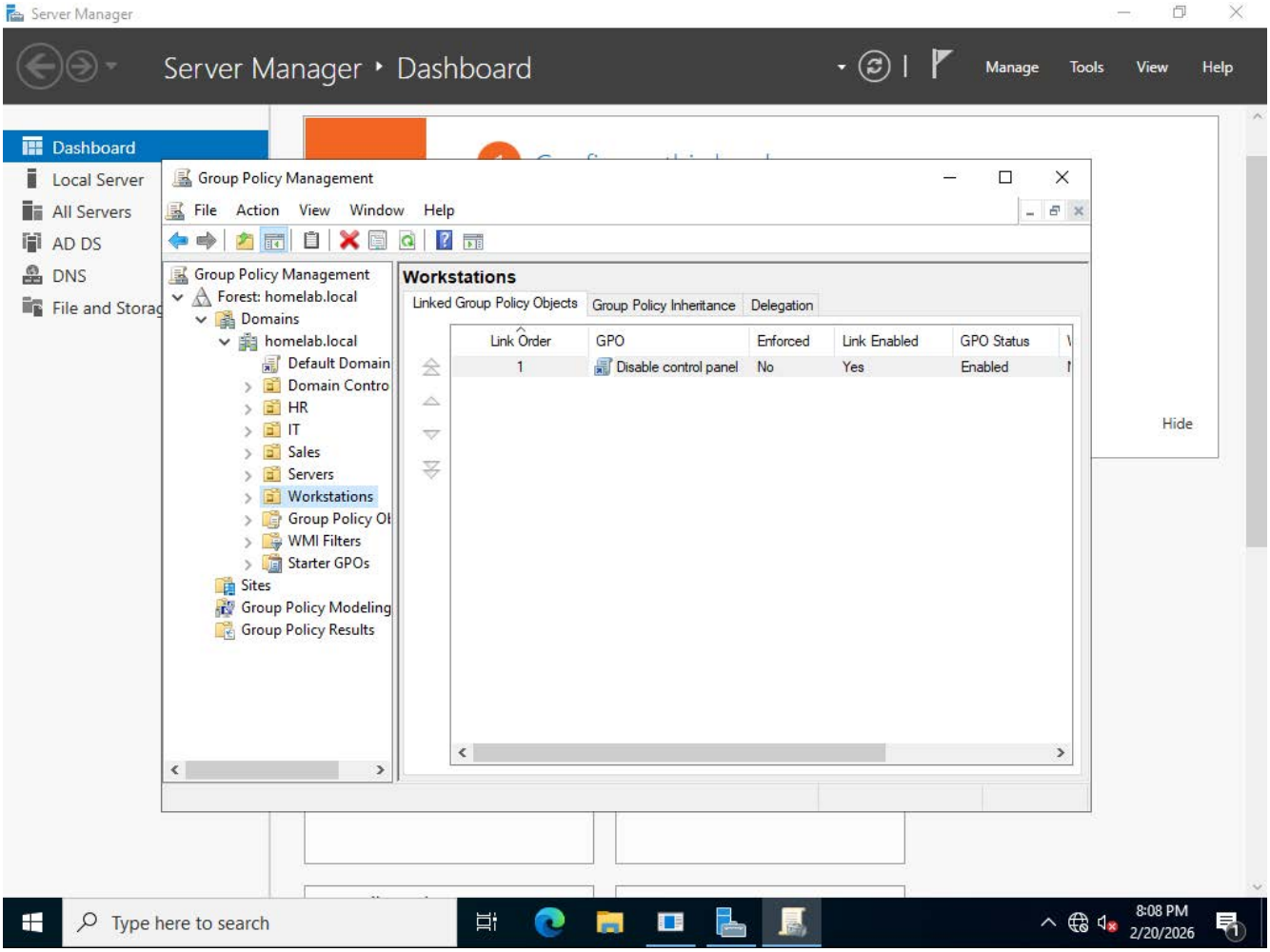
7. Assigned user John to IT OU and Client01 to Workstations OU



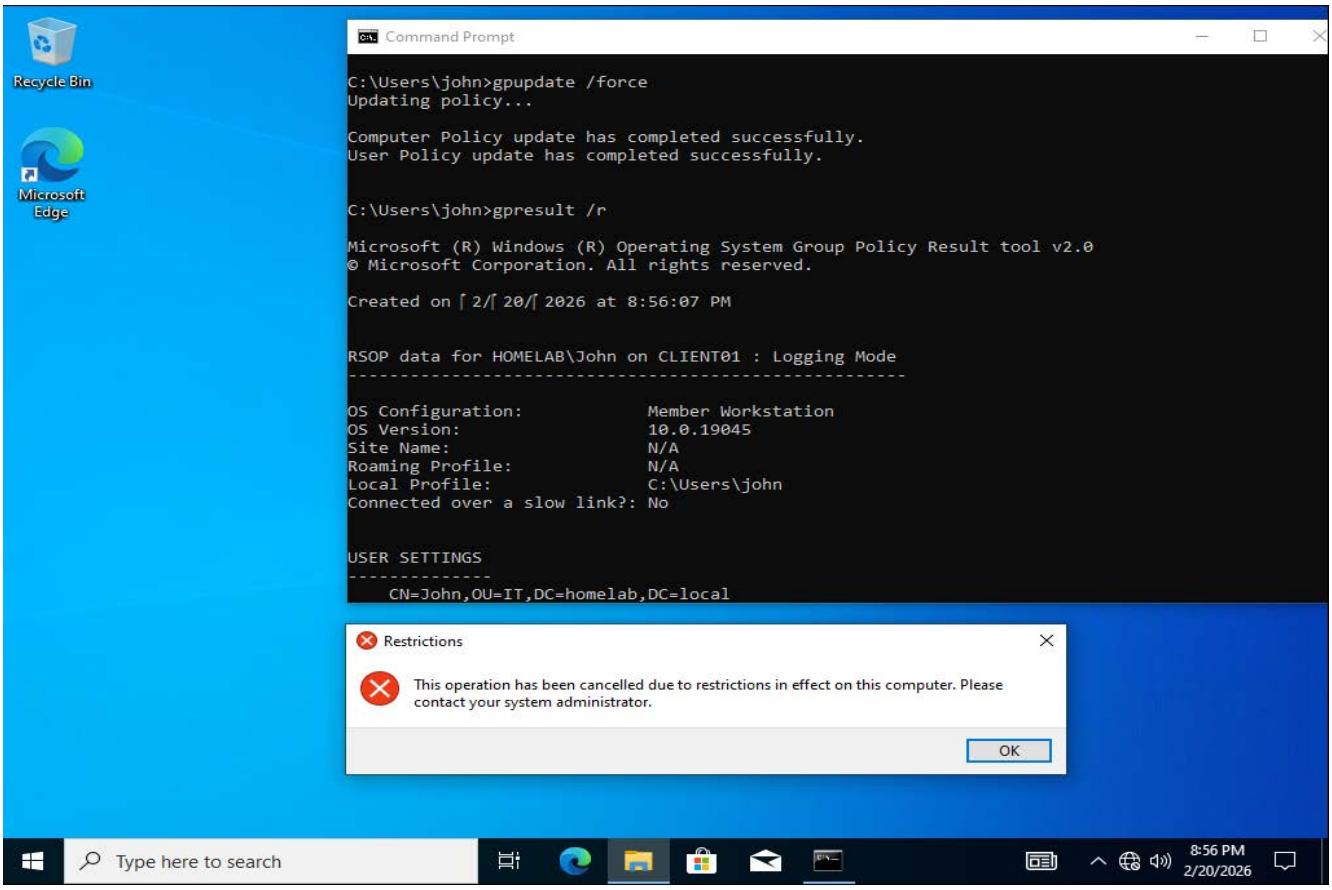
8. Created Security group and added John



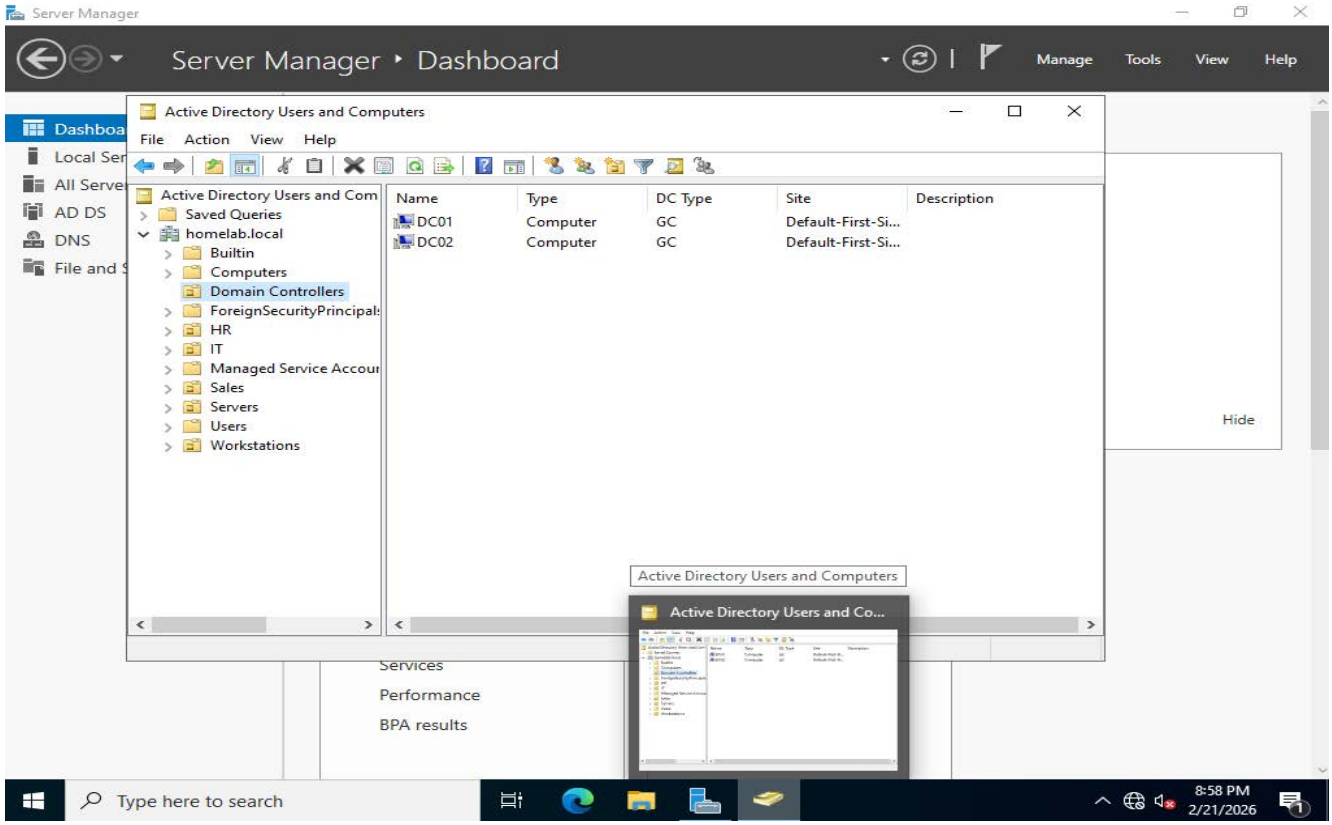
9. Created GPO for Client01 to disable control panel



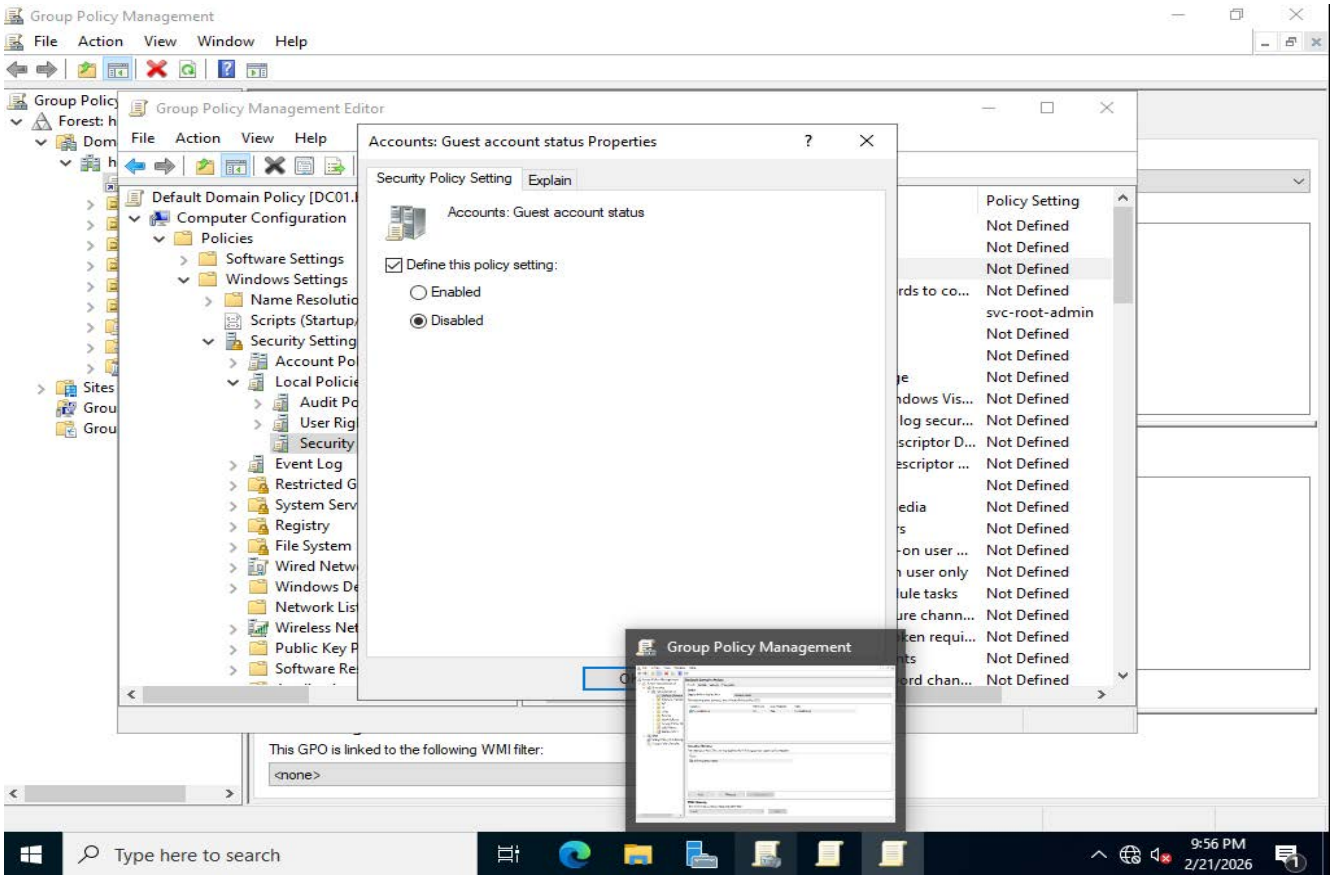
10. Applied and updated Group Policy on client and verified



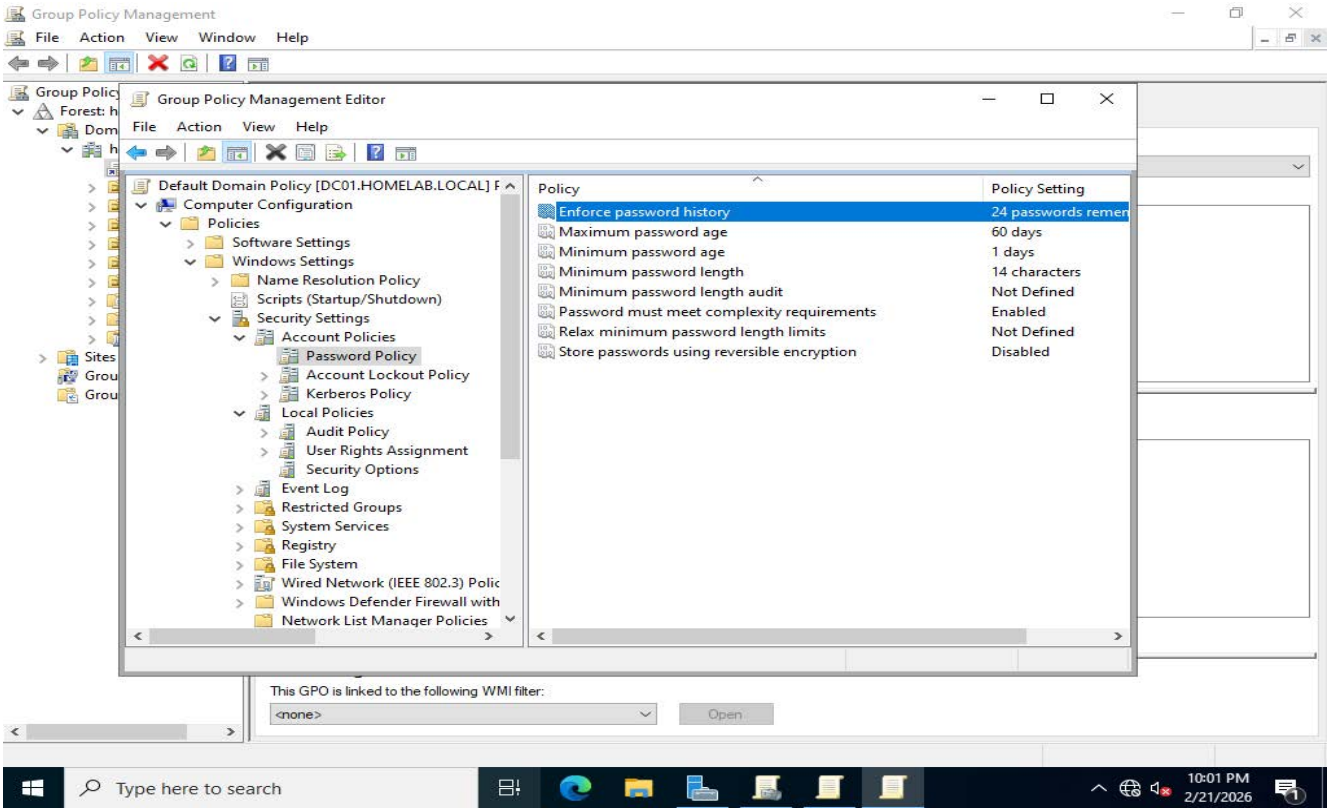
11. Replicated DC01 for fault tolerance, redundancy and high availability



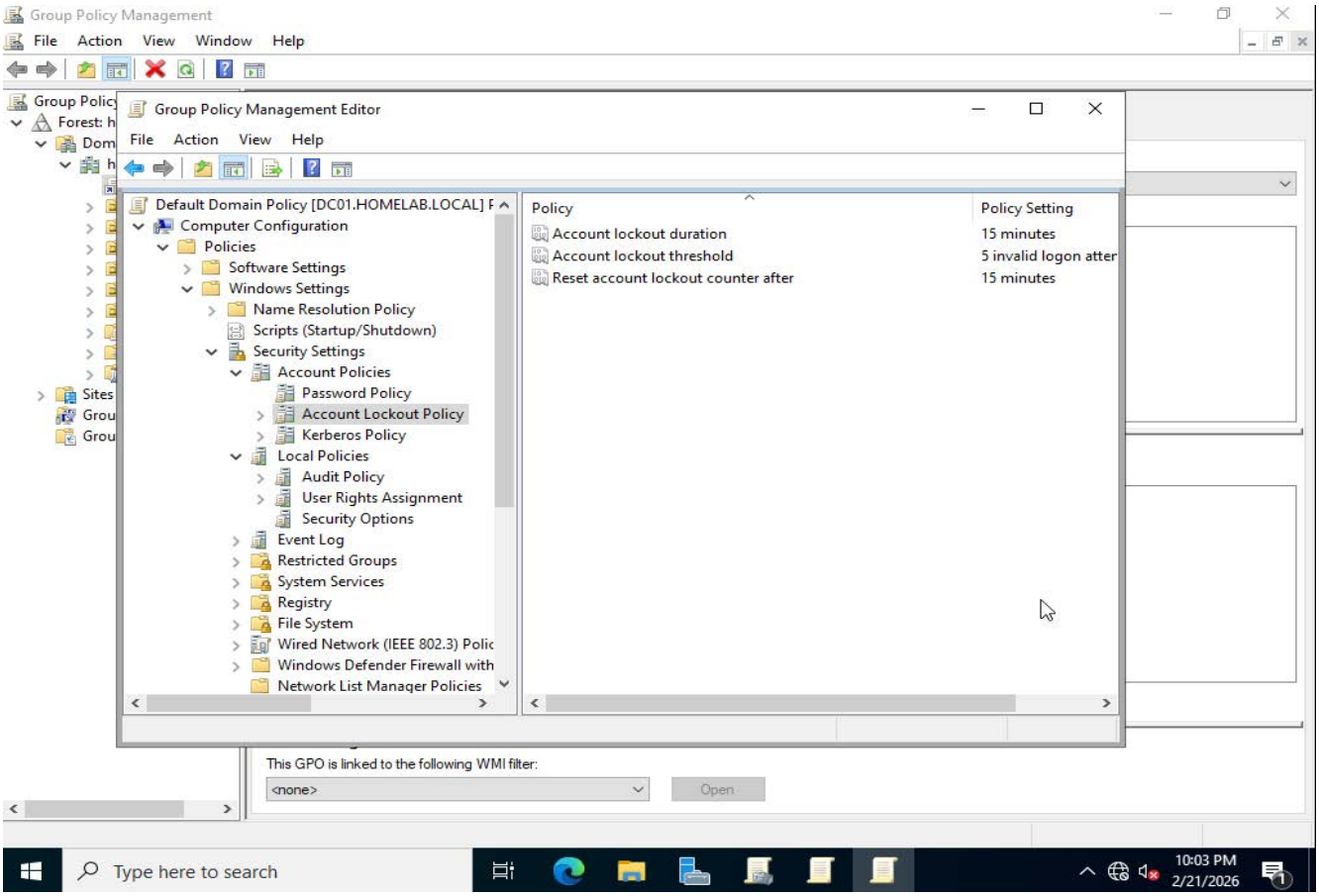
12. Disabling guest accounts/ hardening



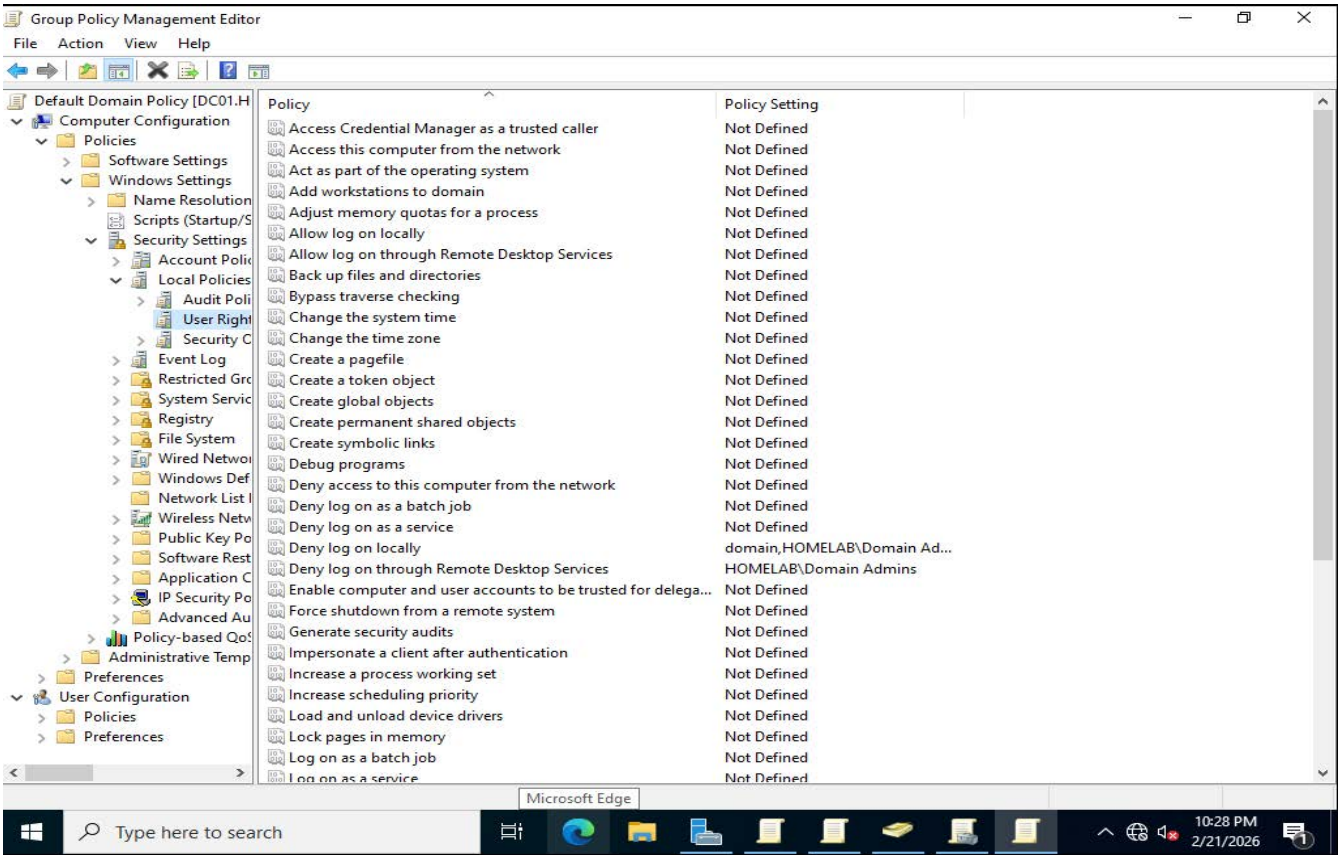
13. Configured and enforced password policy GPO



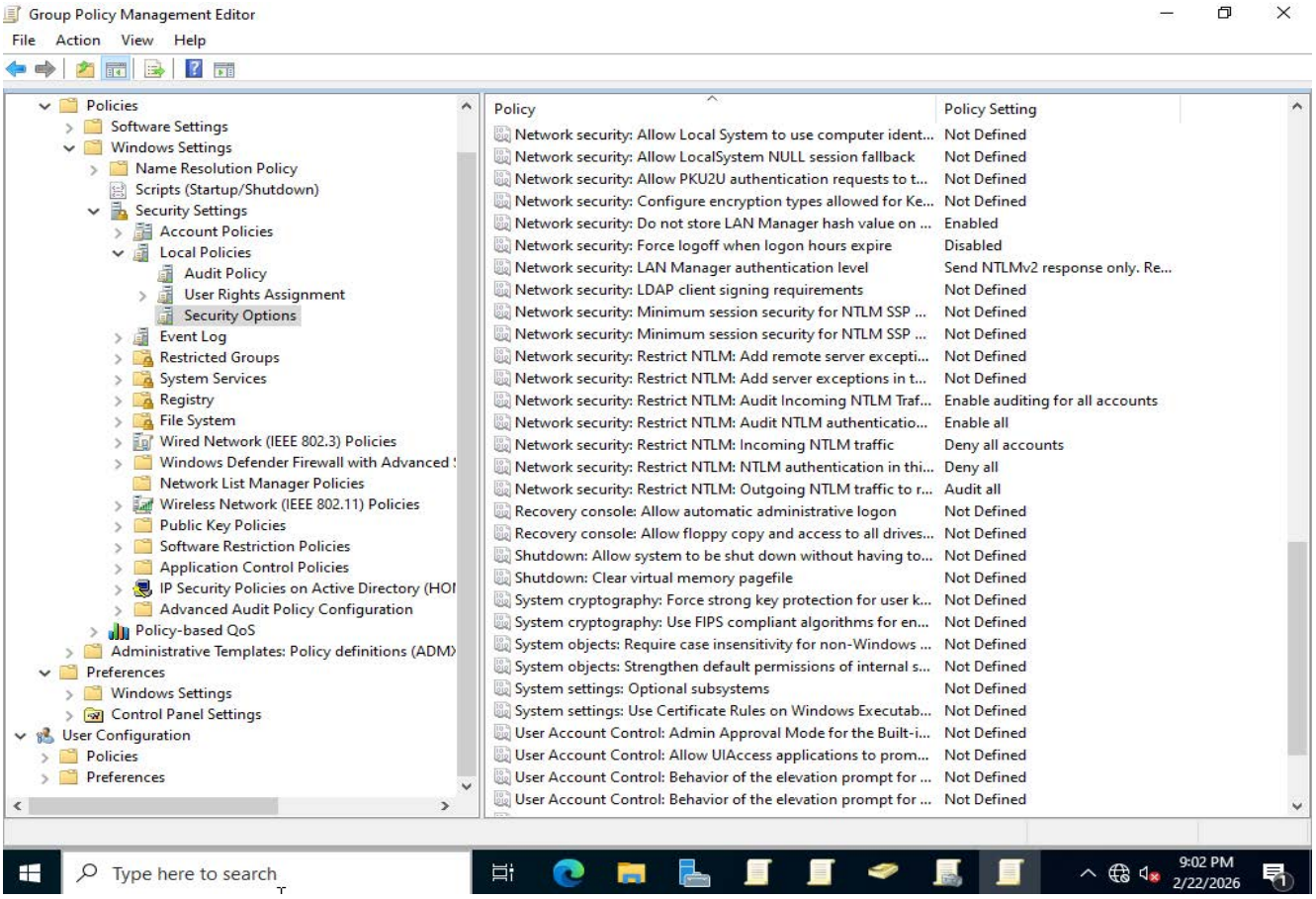
14. Configured and enforced account lockout policy GPO



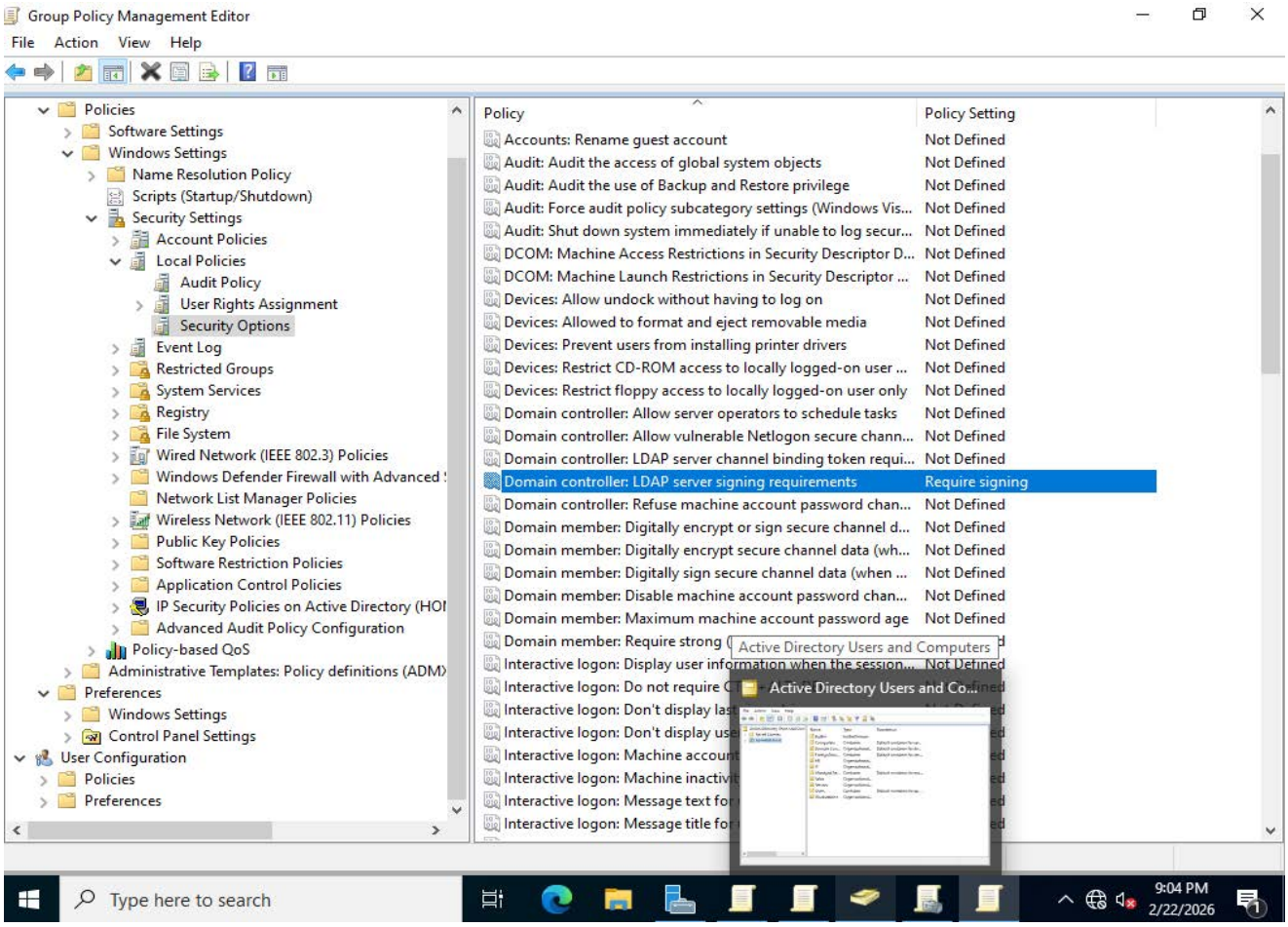
15. Denied admin logon from client workstations



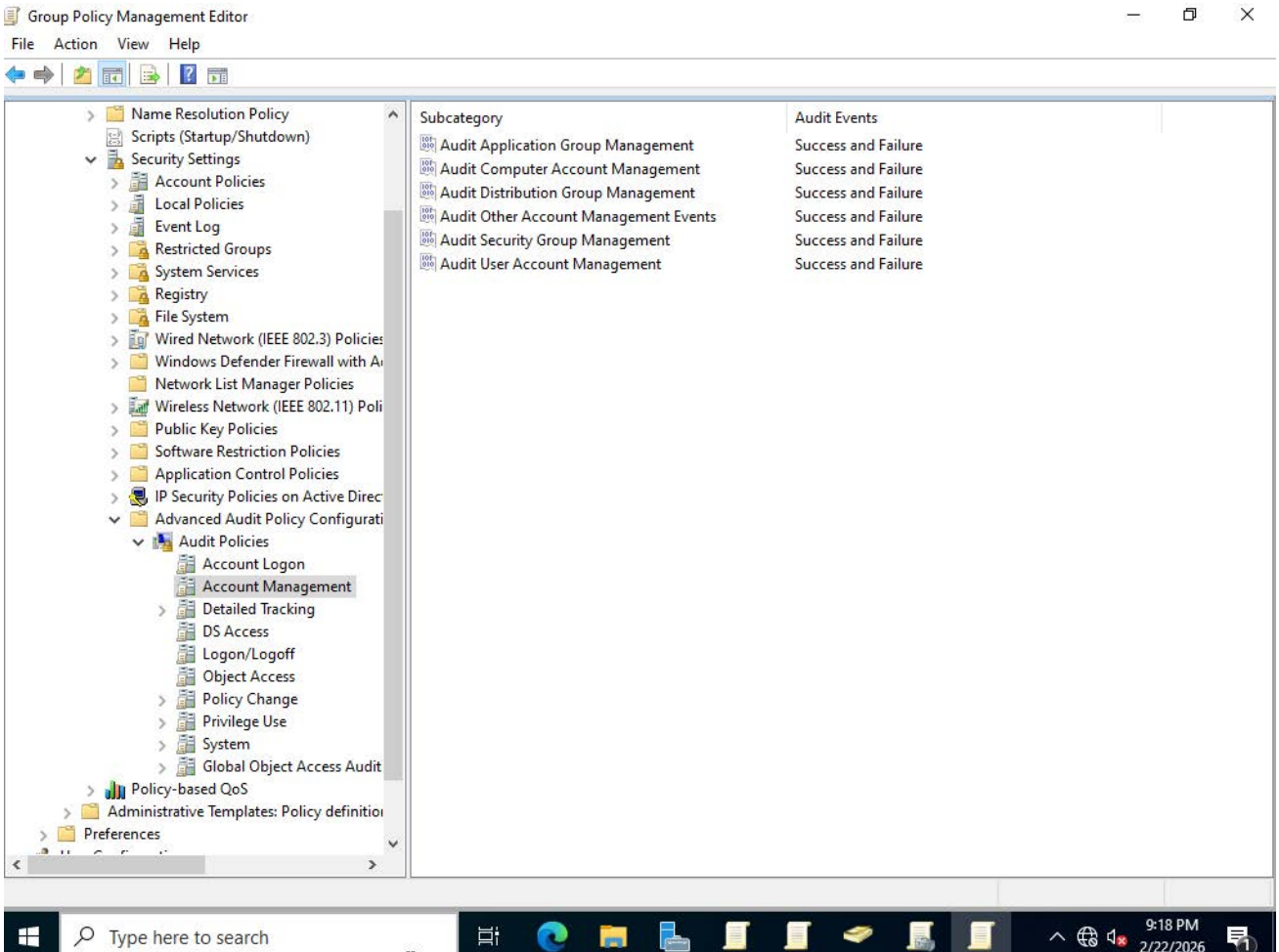
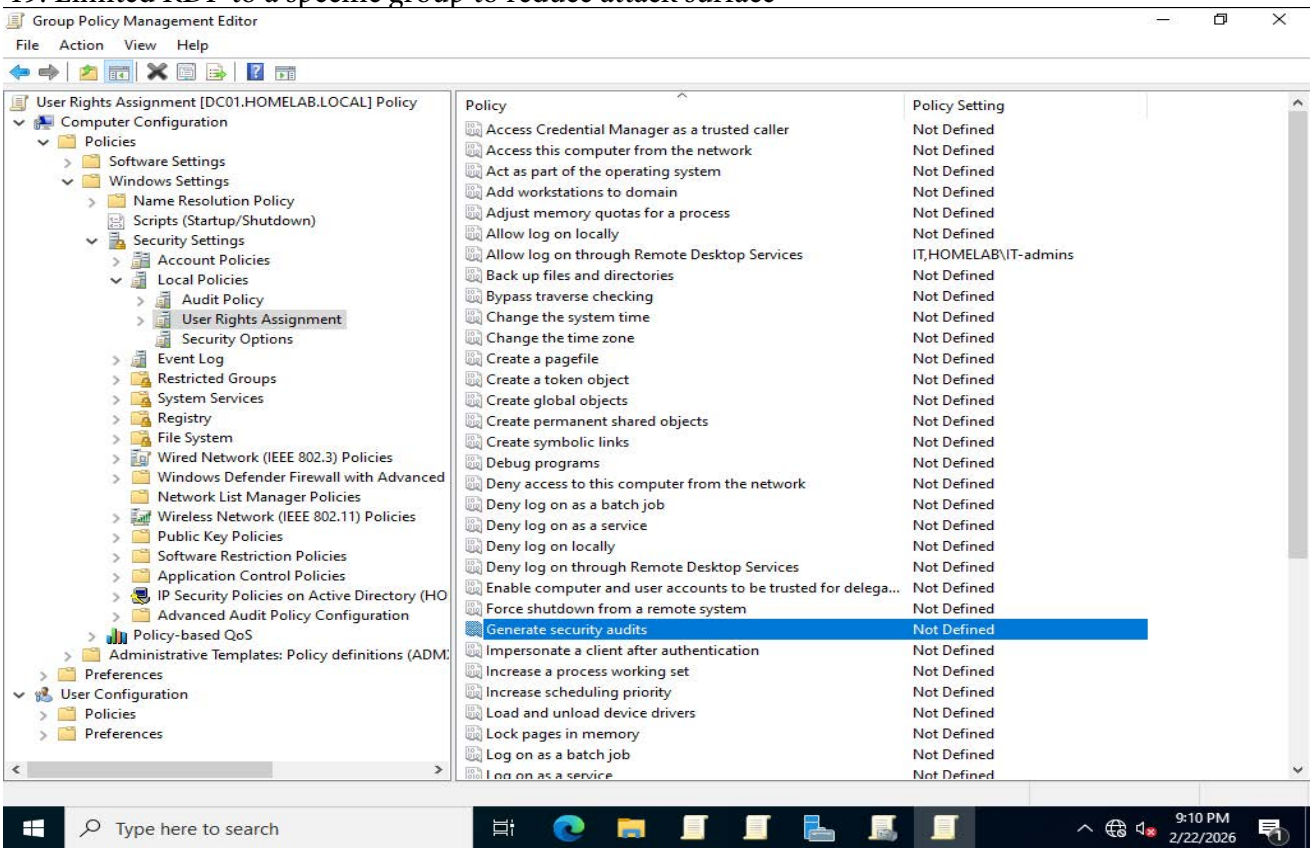
16. Disabled NTLM because it is an old authentication method compared to Kerberos and enabled NTLM auditing for security monitoring.



17. Enabled LDAP signing for directory integrity



18. Enabled Auditing and logging for Logon/ logoff, DS access, Account login, detailed tracking, and account management



19. Limited RDP to a specific group to reduce attack surface